Probabilistic Generative Model for Hyperspectral Unmixing Accounting for Endmember Variability

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***Abstract*— The complex nature of hyperspectral images makes the analysis of spectral signatures a challenging task in remote sensing. For quantitative analysis, spectral unmixing is a well-established and effective tool to analyze the spectra and spatial distribution of substances in the scene. The classical unmixing algorithms usually fail to tackle spectral variability caused by variations in environmental conditions. Many variants based on the linear mixing process have been proposed to tackle this problem; however, the spectral variability modeling capacity of these algorithms is usually insufficient. In this article, we present a probabilistic generative model to address endmember variability and provide more accurate abundance and endmember estimates. The proposed model simultaneously extracts the endmembers and estimates abundances in an unsupervised manner. In particular, it allows fitting arbitrary endmember distributions through the nonlinear modeling capability of neural networks compared to other methods that use parametric endmember variability models. The performance of the proposed approach is evaluated on both synthetic and real datasets. Experimental results show its superiority in comparison with other state-of-the-art methods. The code of this work is available at https://github.com/shuaikaishi/PGMSU for the sake of reproducibility.**

***Index Terms*— Endmember variability, hyperspectral unmixing, probabilistic generative model, variational inference (VI).**

NOTATION

N	Number of pixels.
P	Number of endmembers.
L	Number of spectral bands.
J	Number of dimensions of latent variable.
$\boldsymbol{a}_n \in \mathbb{R}^P$	Abundance vector of the nth pixel.
$\boldsymbol{m}_n \in \mathbb{R}^L$	Endmember vector of the nth pixel.
$\boldsymbol{e}_n \in \mathbb{R}^L$	Additive noise of the nth pixel vector.
$\mathbf{1} \in \mathbb{R}^P$	All-one vector.
$\mathbf{0} \in \mathbb{R}^P$	All-zero vector.
$\boldsymbol{z} \in \mathbb{R}^J$	Latent variable vector.
$\boldsymbol{\mu}_z \in \mathbb{R}^J$	Mean of latent variable.
$\boldsymbol{I}$	Identity matrix.
$\boldsymbol{\Sigma}_z \in \mathbb{R}^{J\times J}$	Covariance of latent variable.
$\boldsymbol{M} \in \mathbb{R}^{L\times P}$	Endmember matrix.
$\boldsymbol{\Lambda} \in \mathbb{R}^{L\times L}$	Noise covariance matrix.
$\mathcal{Y}$	Set of observed pixels.
$\succeq$	Componentwise inequality between vectors.
$\text{vec}(\cdot)$	Vectorize a matrix.
$(\cdot)^{\text{T}}$	Transpose a vector.
$(\cdot)^{(i)}$	ith element of a vector.
$D_{KL}(\cdot\|\|\cdot)$	Kullback–Leibler (KL) divergence between two distributions.
$\odot$	Hadamard product.

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I. INTRODUCTION

HYPERSPECTRAL imaging allows collecting both spatial and spectral information of the observation area. Spectral imaging acquires more intrinsic information about the electromagnetic property of substances than color imaging techniques, and this technique has been extensively used in many fields, such as earth observation [2], astronomy [3], surveillance [4], food processing [5], medical and surgical diagnostics [6] in the recent years. Hyperspectral images generally contain several dozens to hundreds of wavebands. However, they usually present a low spatial resolution due to the tradeoff between energy spread over spatial regions and spectral bands. In particular, in the field of airborne and satellite remote sensing [7], an acceptable signal-to-noise ratio (SNR) requires, for each pixel, a large instantaneous field of view (IFOV) to collect more light. Consequently, a pixel of hyperspectral data can usually include more than one substance, and this spectral mixture greatly impairs the performance of extracting information from the data.

Spectral unmixing is thus one of the most crucial and challenging problems to address in hyperspectral data processing, aiming at obtaining the spectra of pure substances, known as endmembers, and their corresponding proportions, known as abundances, in the image. The simplest but most widely used mixing model of the spectral mixture is the linear mixing model (LMM) [8], which assumes that mixed pixels are linear

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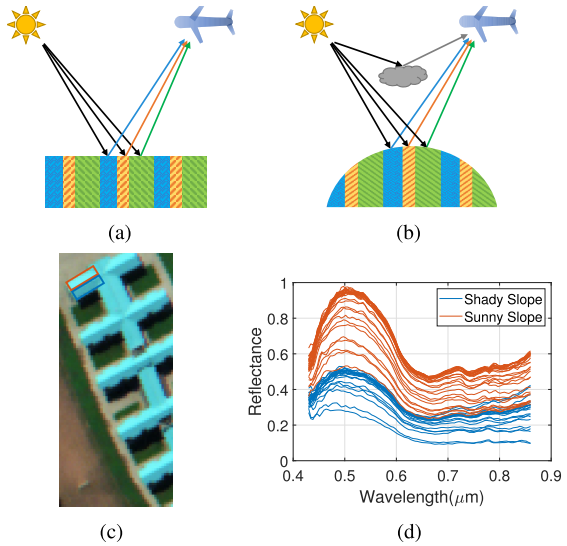


Fig. 1. Illustration of the spectral mixing model and real situation. (a) LMM. (b) Spectral variability caused by environmental conditions. (c) ROI of roof. (d) Endmember spectrum of ROI in the left figure.

combinations of the endmembers weighted by their corresponding surface area ratio, as shown in Fig. 1(a). However, the vanilla LMM cannot represent the spectral variability caused by atmospheric, topographic, temporal effects for instance, as shown in Fig. 1(b). A typical example is provided in Fig. 1(c) and (d), depicting a region of interest (ROI) of the Pavia University [see Fig. 1(c)] and the spectral signatures of the roof [see Fig. 1(d)] under different lighting conditions, which vary considerably. This illustrates how endmember variability makes the unmixing problem in remote sensing more challenging [9].

A. Motivation

Most existing unmixing models that consider endmember variability are based on arbitrarily chosen endmember variation models that are likely to only perform well in suitable scenarios. For example, scaling endmembers or pixels using illumination coefficients, or modeling endmembers with a parametric probability distribution, results in simple, yet restrictive models [10]. Other deep learning-based methods account for endmember variability via nonlinear mixing models, which often lack interpretability and make it difficult to estimate endmembers present at different locations in the scene.

On the other hand, probabilistic generative models combined with deep learning to recognize underlying patterns in data in an unsupervised manner have recently attracted significant attention [11]–[14]. The nonlinear modeling capabilities of neural networks allow such models to far exceed the modeling capabilities of classical statistical distributions based on simpler, parametric models. Therefore, these methods inspire us to employ deep neural networks for identifying the statistical properties of the endmember variability and generate endmember samples to address the blind unmixing task.

B. Methodology Overview and Contributions

In this article, a probabilistic generative model for spectral unmixing (PGMSU) accounting for endmember variability is proposed. PGMSU follows an encoder–decoder network architecture for simultaneously estimating abundances and endmembers. The inference model, that is, the encoder, learns the latent representation of endmember variability, meanwhile estimating the abundances. The generative model, that is, the decoder, maps the latent variables into endmembers and then mixes linearly, using the corresponding abundances to produce estimates of input pixels according to the LMM. Specifically, the main contributions of this article can be summarized as follows.

1) We develop a general two-stream deep learning-based architecture for spectral unmixing to address the endmember variability, which gives pixel-wise endmembers from their latent representation. The proposed model works in an unsupervised manner and is applicable to remote-sensing data without ground truth.
2) Beyond the classical probability distributions proposed previously to model the endmember variability, PGMSU first considers arbitrary endmember distributions. Such distribution is essentially a posterior distribution conditioned on the observed pixels, which is associated with the latent representations of endmembers and will be approximated by variational inference (VI) in this article.
3) We perform the learning of the endmember generative process and the estimation of abundances simultaneously, which is more efficient than two-stage algorithms that first extract endmembers and then estimate abundances.

The remainder of this article is organized as follows. Section II introduces the notation used in the article and describes related works in unmixing considering endmember variability. Section III presents the proposed model and its implementation detail. Experimental results and consequent discussion are presented in Section IV. Finally, a conclusion and future outlook are drawn in Section V.

II. Related Work

In this section, we introduce the LMM and its extension for endmember variability with both the non-probability and probability distribution-based models. Several deep learning-based unmixing models are also briefly described.

A. Linear Mixing Model

Under the LMM assumption, each observed spectrum is a convex combination of "pure material" spectra [8], with the mixing coefficients subject to the so-called abundance sum-to-one constraint (ASC) and abundance nonnegativity constraint (ANC). This combination is classically assumed to be corrupted by additive Gaussian noise. The LMM can be formulated as

$$\mathcal{Y} = \left\{ \boldsymbol{y}_n | \boldsymbol{M}\boldsymbol{a}_n + \boldsymbol{e}_n, \mathbf{1}^{\mathrm{T}}\boldsymbol{a}_n = 1, \boldsymbol{a}_n \succeq \mathbf{0}, \forall n \right\} \tag{1}$$

where $\boldsymbol{M} = [\boldsymbol{m}_1, \boldsymbol{m}_2, \ldots, \boldsymbol{m}_P]$ is the endmember matrix, $\boldsymbol{a}_n$ is abundance vector, and $\boldsymbol{e}_n$ is the noise term for nth pixel. Geometrically, the endmembers are located at the vertices of a simplex because of the convex combination above with

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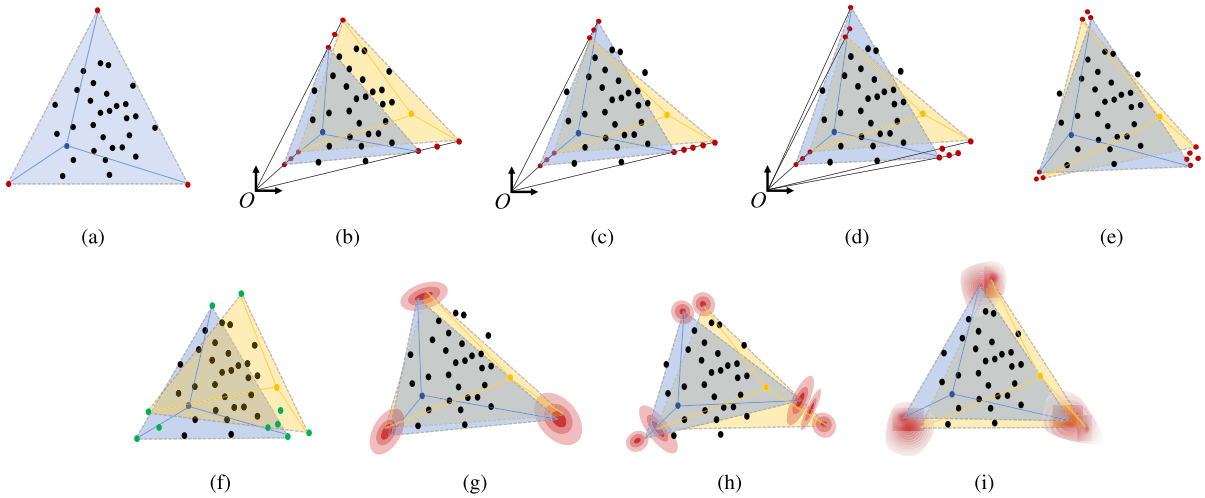


Fig. 2. Illustration of the LMM and its variants for addressing the endmember variability. We have three endmembers represented by the red dots, green dots, or shadow areas. Black dots represent observed pixels. Blue and orange dots mean these points are mixed by vertices corresponding to different color triangles. Green dots represent the available endmember library for MESMA. (a) FCLSU. (b) SCLSU. (c) ELMM. (d) GLMM. (e) PLMM. (f) MESMA. (g) NCM. (h) GMM. (i) PGMSU.

the assumption that endmember vectors are affinely independent [15], as shown in Fig. 2(a). The classical unmixing pipeline consists of two stages: 1) extraction of the vertices of the data simplex as endmembers, that is, using vertex component analysis (VCA) [16] or N-FINDR [17] and 2) estimation of the abundances subject to ASC and ANC using optimization algorithms such as fully constrained least squares (FCLS) [18]. Additional abundance constraints can be introduced to improve performance, such as sparsity [19] and smoothness [20]. Later, non-negative matrix factorization techniques are also introduced to alternatively optimize the abundances and endmembers as a one-stage algorithm [21].

B. Non-Probabilistic Models in LMM Variants

Endmember variability can be caused by a variety of environmental factors, including atmospheric, topographic, and/or temporal effects among others, and its occurrence leads to significant estimation errors throughout the spectral unmixing process [22]. Several extended LMM-based models considering endmember variability have been proposed. In the first class of LMM variants, the statistical distribution of endmembers for each pixel is usually not considered, but parametric endmembers are used instead. These algorithms are referred to as non-probabilistic extensions of the LMM in the next paragraph.

From a practical point of view, it is reasonable to consider the geometric and photometric parameters of the scene to account for endmember variability. Hapke's model [23] is one of the most popular physics-based methods; however, it is very difficult to measure or estimate its numerous physical parameters. Fortunately, Hapke's model can be reasonably well approximated by other flexible and less computationally complex models [24]. A simple approximation method consists of adding a scalar factor to each pixel accounting for the total intensity at each position. The scalar coefficients make the simplices composed of different endmembers geometrically similar, as illustrated by the triangles (for $P = 3$) in Fig. 2(b). The corresponding optimization algorithm is the so-called scaled constrained least-squares unmixing (SCLSU) [25]. More generally, these scalar coefficients can be extended to vectors or matrices to achieve higher flexibility in modeling each endmember or wavelength interval, leading to the extended linear mixing model (ELMM) [26] and generalized linear mixing model (GLMM) [27], respectively. These two models relax the strong simplex constraint in a more flexible way than SCLSU, as shown in Fig. 2(c) and (d).

In addition, a perturbation term can be introduced to compensate for errors in the LMM. A representative model for compensation of endmembers is the perturbed linear mixing model (PLMM) [28] which adds a perturbation term to the endmember matrix, as shown in Fig. 2(e). However, such a perturbation model lacks physical explanation and makes it hard to express the large-scale endmember variability due to changing illumination conditions. A model called augmented linear mixing model (ALMM) [29] was proposed to compensate for the reconstruction error of the LMM, which introduces a spectral variability dictionary to model the mismatch between the observed spectra and its reconstructions. Both GLMM and ALMM are more flexible models than the original LMM. However, these models tend to over-fit the data, leading to biased abundance and endmember estimations.

The multiple endmember spectral mixture analysis (MESMA) [30] algorithm uses a known endmember library and then searches for the endmember and abundance combinations via minimization of reconstruction errors, as shown in Fig. 2(f). However, MESMA may become intractable when the library size is large and its performance strongly depends on the accuracy of the spectral library used to represent the materials of the scene.

C. Probabilistic Models in LMM Variants

Within this second set of methods, endmember variability is modeled by considering the endmembers as random quantities,

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following parametric distributions. The normal compositional model (NCM) [31] introduces Gaussian distributions to model the endmembers of each substance, as shown in Fig. 2(g). With the convex combination of the LMM, the observed pixels follow a new Gaussian distribution with a modified covariance matrix. The abundances and the prior means and covariance matrices of endmembers can be estimated using the expectation-maximization algorithm. There are two main drawbacks in using Gaussian distributions for modeling the endmembers: 1) the Gaussian distribution allows negative samples and 2) the probability density functions of endmembers for each pixel may be multi-modal. There are, however, extensions available to fix the above shortcomings. The beta compositional model (BCM) [32], [33] uses Beta distributions to model endmembers and satisfy the non-negativity constraints, which gives a potentially asymmetric probability density more flexible than the Gaussian model. On the other hand, the Gaussian mixture model (GMM), as shown in Fig. 2(h), considers multiple clusters of endmembers and the problem is solved by generalized expectation-maximization [34], but this algorithm needs to evaluate the precision matrices of the endmembers, which induced a high computational cost with hundreds of bands. Besides, there is another class of models based on the hierarchical Bayesian frameworks, which introduce the prior distributions of abundances and endmembers and construct likelihood according to LMM. Then endmembers are sampled from posterior distribution via the Markov chain Monte Carlo (MCMC)-based method. However, this approach is limited by the prior distributions and may not capture the multimodal distribution of endmembers [35].

D. Related Deep Learning-Based Model for Unmixing

With the booming development of deep neural networks (DNNs) for a variety of imaging problems, deep autoencoders have been introduced for spectral unmixing. Used as an end-to-end method, it has leveraged its power for joint abundance and endmember estimation in the blind spectral unmixing context [36]–[39]. These approaches model the nonlinear mapping from pixels to abundances via an encoder realized by a neural network. The pixel spectra can be reconstructed by a linear decoder [36] (linear unmixing) or a nonlinear decoder [39] (nonlinear unmixing). A deterministic, nonlinear decoder could model the endmember variability, but it would lack a physical interpretation and it is thus limited in considering the diversity of endmembers. Convolutional neural networks (CNNs) were introduced [40]–[42] to leverage spatial information. A CNN allows unmixing of the whole image or a patch of an image as input, whereas the previous fully connected neural networks were only able to perform this task pixel-wise, which is more adapted to large-size images. However, to the best of our knowledge, no CNN-based models for spectral unmixing accounting for endmember variability have been proposed yet. Nevertheless, DNN-based and CNN-based methods give more accurate unmixing results compared to conventional algorithms, thanks to the nonlinear representation capacity of deep neural networks.

Recently, generative models have attracted considerable attention in the deep learning community. Typical models include generative adversarial networks (GANs) [12], variational autoencoders (VAEs) [11], autoregressive models [14], flow-based models [13], and so on. These models, called deep generative models (DGMs), have been successfully applied to learn the underlying structure in high-dimensional data and allow the generation of new samples [43]. Compared to other DGMs, a VAE allows stable training and fast sampling which has been applied to spectral unmixing. Su *et al.* [38] use a stacked autoencoder to remove outliers from the hyperspectral images and followed by a VAE to infer the latent variables referred to as abundances. Besides, the latent variable can also be interpreted as a nonlinear representation of endmembers. Borsoi *et al.* [44] train the VAE to learn the features of the existing spectral library and then generate new endmember samples to augment the library for the MESMA algorithm. However, this method still relies on a known spectral library which may not apply to real data. In [45], the deep generative unmixing algorithm (DeepGUn) trains a VAE with multiple pure pixels extracted from the data and then alternatively optimizes the latent variables and abundances. However, this approach depends on the quantity and quality of the pure pixels extracted, and the parameters of the VAE and abundances are not jointly learned which may lead to suboptimal solutions.

III. Proposed Model

In this section, a deep PGMSU is proposed to tackle endmember variability in the hyperspectral unmixing problem.

A. Generative Model

According to the LMM (1) and considering the different endmembers in each pixel, we have

$$\boldsymbol{y}_n|(\boldsymbol{M}_n, \boldsymbol{a}_n) \sim \mathcal{N}(\boldsymbol{M}_n\boldsymbol{a}_n, \boldsymbol{\Lambda}) \tag{2}$$

where $n = 1, 2, \ldots, N$ and $\boldsymbol{\Lambda} = (\lambda_0 \boldsymbol{I})^{-1}$ is the covariance matrix of the observation noise. Then, we introduce a vector $\boldsymbol{z}_n$ for each pixel which is the latent representation of endmembers of that pixel, encoding their variability. For each pixel, we have

$$\text{vec}(\boldsymbol{M}_n) = f_{\boldsymbol{\theta}}(\boldsymbol{z}_n) \tag{3}$$

where $\text{vec}(\boldsymbol{M}_n) = [\boldsymbol{m}_{1n}^{\mathrm{T}}, \boldsymbol{m}_{2n}^{\mathrm{T}}, \ldots, \boldsymbol{m}_{Pn}^{\mathrm{T}}]^{\mathrm{T}}$ and $f_{\boldsymbol{\theta}}(\cdot)$ is a nonlinear function parameterized by $\boldsymbol{\theta}$ that associates an endmember matrix with its representation. The physical interpretation of $\boldsymbol{z}_n$ could be a quantified value of environmental conditions, including lighting, topography, atmospheric effects, and so on at the current pixel location.

In brief, the generative process of the endmembers can be formulated by (3). Then, the observed pixels can be reconstructed by (2) after optimizing the parameters $\boldsymbol{\theta}$ and inferring the latent variables $\boldsymbol{z}_n$ and abundances $\boldsymbol{a}_n$. The directed acyclic graph of the generative model is illustrated in Fig. 3(a).

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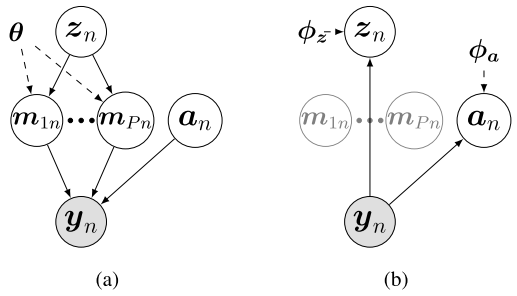


Fig. 3. Directed acyclic graph for the PGMSU. (a) Top-down: Generative model is parameterized by $\boldsymbol{\theta}$ to produce endmembers, $\boldsymbol{m}_{1n}, \ldots, \boldsymbol{m}_{Pn}$, from their latent representation $\boldsymbol{z}_n$. Then endmembers are mixed with their abundances, $\boldsymbol{a}_n$, using the LMM to produce the observed pixel $\boldsymbol{y}_n$. (b) Bottom-up: inference model is a two-stream parameterized by $\boldsymbol{\phi}_z$ and $\boldsymbol{\phi}_a$, respectively, to infer the latent vectors and abundances according to the observed pixel $\boldsymbol{y}_n$.

B. Inference Model

Conventionally, and as discussed in Section II, models that consider the effects of environmental conditions require explicit modeling with physical parameters in the scene, for example, Hapke's model [23] depends on incidence angle and photometric parameters, Kubelk–Munk's model [46] depends on absorbance parameters and the PROSPECT model [47] depends on the biophysical parameters of vegetation. However, it is often not possible to get or estimate some of these parameters accurately and real endmembers are thus inaccessible. From the Bayesian point of view, the true posterior distribution, $p(\boldsymbol{M}_n|\boldsymbol{y}_n)$, is intractable because $p(\boldsymbol{M}_n)$ and $p(\boldsymbol{y}_n)$ are unknown. Fortunately, there are several methods to approximate posterior distributions, such as VI and MCMC. MCMC is widely used in Bayesian inference to draw the samples in the posterior distribution. However, it is difficult to determine whether samplers converge to the target distribution, which results in a long convergence time as most samplers tend to sample in local regions of high probability, resulting in the lack of sample diversity. In this article, the inference is performed via stochastic gradient variational Bayes (SGVB) proposed in [11], which gives a deterministic approximation of the posterior distribution and is scalable for large datasets.

Specifically, we use a neural network $q_{\boldsymbol{\phi}_z}$ parameterized by $\boldsymbol{\phi}_z$ as an approximation of the true posterior distribution of the latent vector $\boldsymbol{z}_n$ conditioned on the pixel $\boldsymbol{y}_n$. The posterior is approximated by a Gaussian as follows:

$$q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n) = \mathcal{N}(\boldsymbol{\mu}_z(\boldsymbol{y}_n), \boldsymbol{\Sigma}_z(\boldsymbol{y}_n)) \tag{4}$$

where the covariance matrix is assumed to be diagonal for convenience, $\boldsymbol{\Sigma}_z = \text{diag}(\boldsymbol{\sigma}_z^2)$.

To infer the abundances, we directly use a nonlinear function that maps pixels to abundances, stated as follows:

$$\boldsymbol{a}_n = f_{\boldsymbol{\phi}_a}(\boldsymbol{y}_n) \tag{5}$$

where $f_{\boldsymbol{\phi}_a}(\cdot)$ is a nonlinear function parameterized by $\boldsymbol{\phi}_a$. The inference model is illustrated in Fig. 3(b).

The resulting processing pipeline, obtained after training, is detailed in Algorithm 1.

Algorithm 1 PGMSU: Global Framework

Input: observed pixels: $\mathcal{Y} = \{\boldsymbol{y}_n, n = 1, 2, \ldots, N\}$;
1 Infer the latent variables, $\boldsymbol{z}_n$, by (4) with reparameterization trick [11];
2 Generate the endmembers, $\widehat{\boldsymbol{M}}_n$ for each pixel using (3);
3 Infer the abundances, $\hat{\boldsymbol{a}}_n$ using (5);
4 Perform the LMM in (2) to reconstruct the input data $\hat{\boldsymbol{y}}_n = \widehat{\boldsymbol{M}}_n \hat{\boldsymbol{a}}_n$;
Output: estimations of abundance maps, endmembers and reconstructions for input pixels: $\hat{\boldsymbol{a}}_n, \hat{\boldsymbol{m}}_n$ and $\hat{\boldsymbol{y}}_n$, $\forall n$.

C. Model Architecture

This section presents the construction of the generative and inference models using neural networks.

We first model the variational distribution (4), which is constrained to be Gaussian with a diagonal covariance matrix. The mean vector and the diagonal of the covariance matrix of this distribution are modeled by fully connected neural networks (FCNN) with pixels as its inputs, as shown in the upper left part of Fig. 4. We use the classical reparameterization trick [11] to infer the latent representation of the endmembers, which contains two stages: 1) we draw a sample from a standard multivariate Gaussian distribution and 2) the sample is scaled and offset by the standard deviations and means of the variational distribution, instead of directly drawing sample from (4). Endmembers are generated by (3), which is realized by another FCNN. The mapping $f_{\boldsymbol{\theta}}(\boldsymbol{z}_n)$ includes the sigmoid function after the last layer, limiting the generated endmembers to (0, 1), as shown in the right part in Fig. 4. the sigmoid function is given by

$$\text{Sigmoid: } \boldsymbol{x} \leftarrow \frac{1}{1 + \exp(-\boldsymbol{x})} \tag{6}$$

where $\boldsymbol{x}$ represents an arbitrary vector. Abundances are estimated by the branch FCNN, $f_{\boldsymbol{\phi}_a}(\boldsymbol{y}_n)$, which uses Softmax layer as output to ensure the ASC and ANC, as shown in the lower left part in Fig. 4. The softmax layer means that the following element-wise transformation is used:

$$\text{Softmax: } \boldsymbol{x} \leftarrow \frac{\exp(\boldsymbol{x})}{\sum_i \exp(\boldsymbol{x})} \tag{7}$$

where $\boldsymbol{x}$ represents an arbitrary vector. Last, the pixel spectra are reconstructed by the sum of the endmembers, weighted by their corresponding abundances, as shown in the right part of Fig. 4.

Batch normalization (BN) layers [48] are added into our network to speed up the training process and we use the Leaky ReLU nonlinear activation function in the neural network. BN with Leaky ReLU is a general setup in recent neural network architectures [39]. The detailed configuration of PGMSU is shown in Table I.

D. Objective Function/Training Stage

Considering the generative model and the inference model, the training stage of the proposed method consists of

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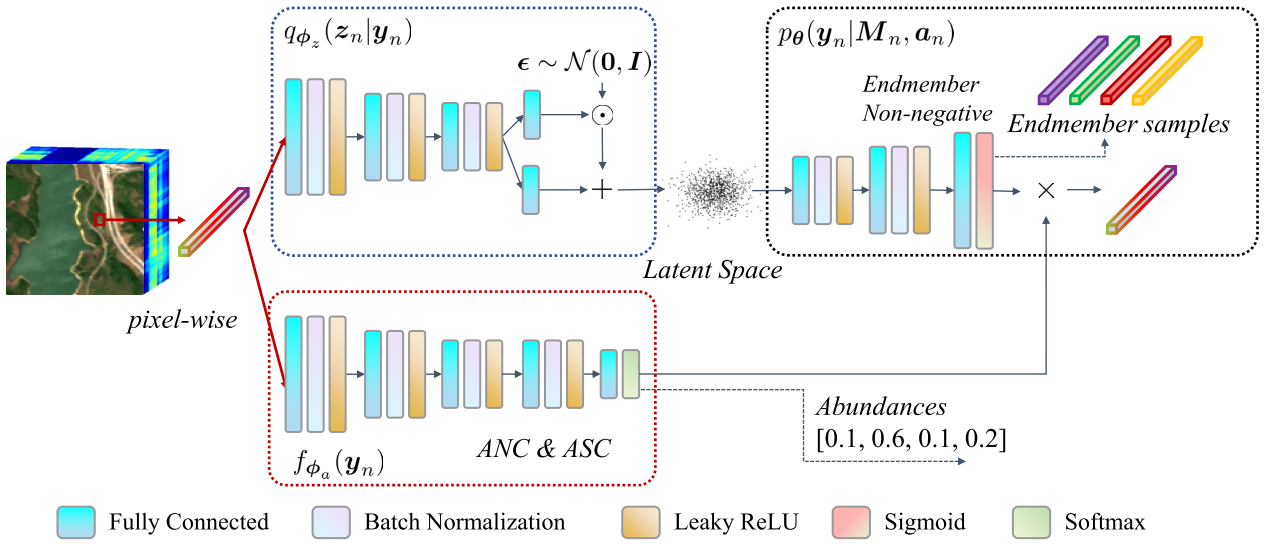


Fig. 4. Overview of the proposed PGMSU architecture. It is a two-stream network, including a VAE as backbone extracting endmembers and another branch estimating abundances.

TABLE I

NEURAL NETWORK CONFIGURATION OF PGMSU. FC AND BN MEAN FULLY CONNECTED LAYER AND BATCH NORMALIZATION LAYER, RESPECTIVELY. LRELU MEANS LEAKY RELU NONLINEAR ACTIVATION FUNCTION

Backbone		Branch	
Layers	neurons	Layers	neurons
FC1 - BN1 - LReLU	$32P$	FC9 - BN9 - LReLU	$32P$
FC2 - BN2 - LReLU	$16P$	FC10 - BN10 - LReLU	$16P$
FC3 - BN3 - LReLU	$4P$	FC11 - BN11 - LReLU	$4P$
FC4 FC5	J J	FC12 - BN12-LReLU	$4P$
FC6 - BN6 - LReLU	$16P$	FC13 - Softmax	P
FC7 - BN7 - LReLU	$64P$		
FC8 - Sigmoid	LP		

estimating the parameters $\{\boldsymbol{\theta}, \boldsymbol{\phi}_a, \boldsymbol{\phi}_z\}$ from a set of observed pixels $\mathcal{Y}$ (without ground-truth abundances or endmembers).

This can be achieved by maximum likelihood estimation (MLE) [11]. The data log-likelihood function is given by

$$\sum_{n=1}^{N} \log p(\boldsymbol{y}_n) = \sum_{n=1}^{N} \mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}_a, \boldsymbol{\phi}_z; \boldsymbol{y}_n) + \sum_{n=1}^{N} D_{KL}(q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)||p(\boldsymbol{z}_n|\boldsymbol{y}_n)) \quad (8)$$

where

$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}_a, \boldsymbol{\phi}_z; \boldsymbol{y}_n) = \mathbb{E}_{q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)}\left[\log p_{\boldsymbol{\theta}}(\boldsymbol{y}_n|\boldsymbol{z}_n)\right] - D_{KL}(q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)||p(\boldsymbol{z})). \quad (9)$$

See Appendix A for more detailed derivations of these expressions. Due to the non-negativity of the KL divergence, the first term on the R.H.S. of (8) is always lower than the log-likelihood function, so it is called the evidence lower bound (ELBO). Once the optimization is done, the variational distribution $q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)$ will be close to the true posterior $p(\boldsymbol{z}_n|\boldsymbol{y}_n)$ that describes the endmember variability conditioned on observed pixels. The density in $p(\boldsymbol{z})$ in (9) is a prior distribution, usually selected as standard Gaussian, that is, $p(\boldsymbol{z}) = \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$. We maximize the ELBO to implement MLE, then we focus on the ELBO.

Furthermore, the expectation in (9) can be approximated by single Monte Carlo sampling according to [11]

$$\mathbb{E}_{q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)}\left[\log p_{\boldsymbol{\theta}}(\boldsymbol{y}_n|\boldsymbol{z}_n)\right] \approx \log p_{\boldsymbol{\theta}}(\boldsymbol{y}_n|\boldsymbol{z}_n) \quad (10)$$

$$= \log p_{\boldsymbol{\theta}}(\boldsymbol{y}_n|\boldsymbol{M}_n). \quad (11)$$

Note that the approximation from line 1 to line 2 comes from sampling only one sample from $q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)$ to compute the expectation. Because the mapping from $\boldsymbol{z}_n$ to $\boldsymbol{M}_n$ is deterministic, assumed by (3), we have

$$p_{\boldsymbol{\theta}}(\boldsymbol{M}_n|\boldsymbol{z}_n) = \begin{cases} 1, & \text{vec}(\boldsymbol{M}_n) = f_{\boldsymbol{\theta}}(\boldsymbol{z}_n) \\ 0, & \text{otherwise.} \end{cases} \quad (12)$$

To reconstruct the input data, $\boldsymbol{a}_n$ is seen as a known value inferred by (5) and added into (11) to perform LMM

$$\log p_{\boldsymbol{\theta}}(\boldsymbol{y}_n|\boldsymbol{M}_n) \approx \log p_{\boldsymbol{\theta}}(\boldsymbol{y}_n|\boldsymbol{M}_n, \boldsymbol{a}_n). \quad (13)$$

Moreover, (13) can be derived as a square error between a pixel and its reconstruction due to the Gaussian noise assumption in (2). More details should refer to Appendix B. This term is called the reconstruction error and denoted as follows:

$$\log p_{\boldsymbol{\theta}}(\boldsymbol{y}_n|\boldsymbol{M}_n, \boldsymbol{a}_n) \propto -\lambda_0||\boldsymbol{y}_n - \boldsymbol{M}_n\boldsymbol{a}_n||^2 = -\lambda_0 \mathcal{L}_{\text{rec}}. \quad (14)$$

The KL divergence in (9) is tractable as it measures the similarity between the Gaussian distribution in (4) and the

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standard Gaussian distribution. Indeed, this term is given by

$$\mathcal{L}_{\mathrm{KL}} = D_{\mathrm{KL}}(q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)||p(\boldsymbol{z})) = \sum_{i=1}^{J} -\frac{1}{2}\left[\log \boldsymbol{\Sigma}_z^{(i,i)} - \boldsymbol{\mu}_z^{(i)2} - \boldsymbol{\Sigma}_z^{(i,i)2} + 1\right] \quad (15)$$

where J is the dimension of the latent space. This term is denoted as $\mathcal{L}_{\mathrm{KL}}$ and we give a short derivation of its expression in Appendix C.

Combining (9)–(15), the objective function of the probabilistic generative model is

$$\mathcal{L}_{\mathrm{PGM}}(\boldsymbol{\theta}, \boldsymbol{\phi}_a, \boldsymbol{\phi}_z) = \lambda_0 \mathcal{L}_{\mathrm{rec}} + \mathcal{L}_{\mathrm{KL}}. \quad (16)$$

Maximizing the ELBO is equivalent to minimizing the objective function above.

Besides, several expected properties can be introduced into the PGMSU as regularized terms. Taking into account the property of ELMM, first, the endmembers of each individual material are expected to present similar shapes, the main effect of variability changing in overall intensity. Thus, we constrain the spectral angle distance (SAD) between the endmembers and their means, which can be formulated as follows:

$$\mathcal{L}_{\mathrm{SAD}} = \frac{1}{NP} \sum_{j=1}^{P} \sum_{n=1}^{N} \arccos \frac{\overline{\boldsymbol{m}}_j^{\mathrm{T}} \boldsymbol{m}_{nj}}{\|\overline{\boldsymbol{m}}_j\| \|\boldsymbol{m}_{nj}\|} \quad (17)$$

where $\overline{\boldsymbol{m}}_j = (1/N) \sum_{n=1}^{N} \boldsymbol{m}_{nj}$.

Second, the minimum volume constraint (MVC) [49] applied to endmembers has already shown to be promising for spectral unmixing. To promote similar endmember properties (i.e., estimated endmembers close to the observed data), we use the endmembers variance [50] as a simpler regularizer, given by

$$\mathcal{L}_{\mathrm{Vol}} = \frac{1}{NLP} \sum_{n=1}^{N} \sum_{j=1}^{P} \left\| \boldsymbol{m}_{nj} - \frac{1}{P} \sum_{j=1}^{P} \boldsymbol{m}_{nj} \right\|^2. \quad (18)$$

Finally, the overall loss of PGMSU becomes

$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}_a, \boldsymbol{\phi}_z) = \mathcal{L}_{\mathrm{rec}} + \lambda_1 \mathcal{L}_{\mathrm{KL}} + \lambda_2 \mathcal{L}_{\mathrm{SAD}} + \lambda_3 \mathcal{L}_{\mathrm{Vol}} \quad (19)$$

where $\lambda_1 \equiv 1/\lambda_0, \lambda_2, \lambda_3$ are hyperparameters.

E. Optimization

For stabilizing the unmixing result and accelerate the training process, a pre-training phase is adopted before training. Specifically, we use the endmembers extracted by VCA to constrain the generative model. The pre-training loss is as follows:

$$\mathcal{L}_{\mathrm{Pre}}(\boldsymbol{\theta}, \boldsymbol{\phi}_a, \boldsymbol{\phi}_z) = \mathcal{L}_{\mathrm{rec}} + \lambda_1 \mathcal{L}_{\mathrm{KL}} + \eta \mathcal{L}_{\mathrm{VCA}} \quad (20)$$

where η is a hyperparameter and

$$\mathcal{L}_{\mathrm{VCA}} = \frac{1}{N} \sum_{n=1}^{N} \|\boldsymbol{M}_n - \boldsymbol{M}_{\mathrm{VCA}}\|_F^2. \quad (21)$$

Overall, the learning strategy of PGMSU is summarized in Algorithm 2.

Algorithm 2 PGMSU: Global Algorithm

Input: observed pixels: $\mathcal{Y} = \{\boldsymbol{y}_n, n = 1, 2, \ldots, N\}$;
hyperparameters: $\lambda_1, \lambda_2, \lambda_3, \eta$;
mini-batch size D, numbers of batches N/D ;
epochs: $maxIter = 200$;
learning rate: $\alpha = 0.001$, $t = 0$.

Output: abundance maps and endmembers estimation: $\boldsymbol{a}_n, \boldsymbol{m}_n, \ \forall n$

```
Initialize the model parameters: Θ = {θ, φ_a, φ_z};
Training stage:
while t ≤ maxIter do
    i = 0;
    t = t + 1;
    for i ≤ N/D do
        Sample D samples from 𝒴;
        Forward propagation:
            Inference model: q_{φ_z}(z_n|y_n);
            Branch network: a_n = f_{φ_a}(y_n);
            Generative model:
                vec(M_n) = f_θ(z_n),
                p_θ(y_n|M_n, a_n);
            Calculate loss 𝓛_Pre(Θ) or 𝓛(Θ)
        Back propagation:
            Calculate gradient ∂𝓛_Pre(Θ)/∂Θ or ∂𝓛(Θ)/∂Θ
        Updata Θ using the Adam optimizer;
        i = i + 1.
    end
end
Test stage:
    Forward propagation: Feed the PGMSU 𝒴;
        Obtain m_n, ∀n by the generative model;
        Obtain a_n, ∀n by the branch network.
```

Our approach is implemented on the Pytorch framework using the mini-batch stochastic gradient descent to learn the model parameters, $\{\boldsymbol{\theta}, \boldsymbol{\phi}_a, \boldsymbol{\phi}_z\}$, which are initialized by random sampling from Xavier uniform distribution [51]. We adopt the Adam [52] optimizer and the learning rate is set to 0.001. Both the pre-training and training phase last 100 epochs with the mini-batch size setting to $N/10$. The dimensionality of the latent variable is setting to 4 and the effect of J on unmixing performance will be analyzed in Section IV. For all experiments, we set $\eta = 0.1$. The details of stochastic back propagation can refer to [53].

One point we would like to mention here is that, if there is more endmembers or richer endmember variability, the posterior distribution, $p(\boldsymbol{M}_n|\boldsymbol{y}_n)$, will become more complex, which challenges the expressiveness of our proposed model and needs to increase the dimensionality of latent vector or increase the number of layers of the generative model to cover the more complex and higher-dimensional distribution of endmembers.

IV. Experiments

In this section, we present experimental results to evaluate the proposed approach both on synthetic and real datasets. The

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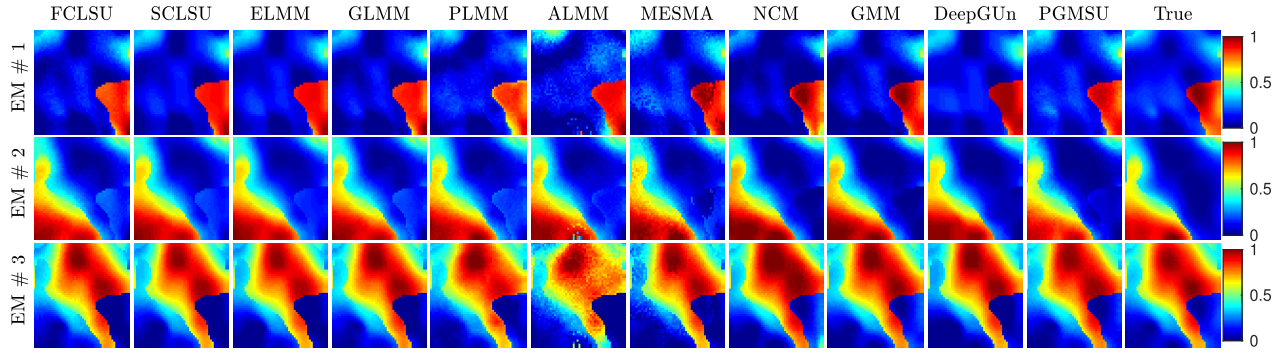


Fig. 5. Abundance maps of unmixing results for different methods on synthetic data. EM #1, EM #2, and EM #3 represent three different materials.

algorithms used for comparison include conventional methods, for example, FCLSU [18], SCLSU [25], MESMA [30], and NCM [31], and state-of-the-art methods, ELMM [26], GLMM [27], PLMM [28], ALMM [29], and GMM [34]. For fair comparisons, all the methods considered are LMM-based, considering endmember variability, except for FCLSU. The algorithm FCLSU is listed here as the baseline to demonstrate the impact of the endmember variability. We constrain the scaling factor of ELMM to be constant at each pixel position to achieve SCLSU. The reference endmembers are extracted by VCA [16], while the endmember library used in MESMA is extracted from the scene using an automated endmember extraction strategy [54]. We constrain the number of Gaussian components for each endmember in GMM to be 1 to achieve the NCM.

In order to evaluate the performance of the different algorithms, we consider the commonly used average root mean square error (aRMSE) for the abundances and endmembers and average spectral angle distance (aSAD) for the endmembers only. When the ground truth of abundance maps and endmembers are available, the $\text{aRMSE}_{\boldsymbol{A}}$ metric is defined as

$$\text{aRMSE}_{\boldsymbol{A}} = \frac{1}{N}\sum_{n=1}^{N}\sqrt{\frac{1}{P}\|\boldsymbol{a}_n - \hat{\boldsymbol{a}}_n\|^2}. \tag{22}$$

For the endmembers, we use

$$\text{aRMSE}_{\boldsymbol{M}} = \frac{1}{NP}\sum_{n=1}^{N}\sum_{j=1}^{P}\sqrt{\frac{1}{L}\left\|\boldsymbol{m}_{nj} - \hat{\boldsymbol{m}}_{nj}\right\|^2} \tag{23}$$

$$\text{aSAD}_{\boldsymbol{M}} = \frac{1}{NP}\sum_{n=1}^{N}\sum_{j=1}^{P}\arccos\frac{\boldsymbol{m}_{nj}^{\mathrm{T}}\hat{\boldsymbol{m}}_{nj}}{\|\boldsymbol{m}_{nj}\|\|\hat{\boldsymbol{m}}_{nj}\|}. \tag{24}$$

If no ground-truth abundance or endmember is available, for example, for most real datasets, we consider the pixel reconstruction error as the main criterion

$$\text{aRMSE}_{\boldsymbol{Y}} = \frac{1}{N}\sum_{n=1}^{N}\sum_{j=1}^{P}\sqrt{\frac{1}{L}\left\|\boldsymbol{y}_n - \hat{\boldsymbol{y}}_n\right\|^2} \tag{25}$$

together with the SAD

$$\text{aSAD}_{\boldsymbol{Y}} = \frac{1}{N}\sum_{n=1}^{N}\arccos\frac{\boldsymbol{y}_n^{\mathrm{T}}\hat{\boldsymbol{y}}_n}{\|\boldsymbol{y}_n\|\|\hat{\boldsymbol{y}}_n\|}. \tag{26}$$

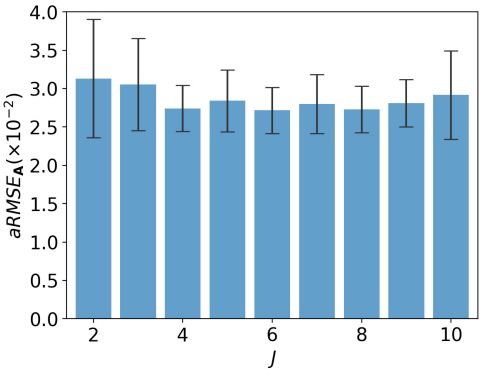


Fig. 6. Abundance aRMSE as a function of the latent space dimension J for the synthetic dataset.

Note that minimizing the reconstruction error is not our main goal and it does not ensure the unmixing task is performed successfully [27], [29].

A. Experiments on the Synthetic Dataset

First, we use a synthetic dataset created in [45] and which consists of three 224-band reference endmembers from the United States Geological Survey (USGS) spectral library [55] and the variability model of GLMM is used. This dataset has 50 by 50 pixels. Endmember scaling factors are smooth both across the spatial and spectral dimensions. True abundance maps are designed to be spatially correlated as shown in the rightmost column of Fig. 5. For this dataset, we set the hyperparameters using $\lambda_1 = 0.1$, $\lambda_2 = 0$, and $\lambda_3 = 0.05$. The other algorithms are implemented by the open source code of the corresponding articles. For synthetic data, the hyperparameters of ELMM, GLMM, PLMM, and DeepGUn are set as in [45]. The hyperparameters of remaining algorithms are searched with the ground truth of abundances.

A key hyperparameter in the design of the PGMSU is the dimensionality J of the latent space. A tradeoff exists between the model capacity and optimization issues. A large value of J enables the generative model to capture a richer pattern of endmember variability and generate more endmembers for different materials, at the cost of higher model complexity and difficulty in the optimization process. To illustrate this, we train the model with different values of J on the synthetic

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TABLE II

QUANTITATIVE PERFORMANCE COMPARISON OF THE TESTED METHODS ON THE SYNTHETIC DATA. THE STANDARD DEVIATIONS OF ALL INDICATORS ARE THE RESULTS OF TEN INDEPENDENT RUNS. THE BEST ONE IS SHOWN IN BOLD, AND THE SECOND BEST ONE IS SHOWN IN BLUE. RUN-TIME (IN SECONDS) IS THE AVERAGE ELAPSED TIME OF EACH METHOD

Methods	aRMSE$_{\boldsymbol{A}}$(10^{-2})	aRMSE$_{\boldsymbol{M}}$(10^{-2})	aSAD$_{\boldsymbol{M}}$(10^{-2})	aRMSE$_{\boldsymbol{Y}}$(10^{-2})	aSAD$_{\boldsymbol{Y}}$(10^{-2})	Run-time(s)
VCA + FCLSU	4.96 ± 0.17	3.47 ± 0.04	3.78 ± 0.16	2.51 ± 0.07	3.78 ± 0.16	0.53
SCLSU	4.32 ± 0.22	3.22 ± 0.14	3.74 ± 0.10	1.51 ± 0.05	2.34 ± 0.09	37.75
ELMM	4.31 ± 0.14	3.08 ± 0.07	3.74 ± 0.10	1.50 ± 0.05	2.34 ± 0.08	33.60
GLMM	4.38 ± 0.23	2.87 ± 0.05	3.56 ± 0.02	1.46 ± 0.01	2.26 ± 0.01	39.08
PLMM	4.62 ± 0.10	2.67 ± 0.03	3.73 ± 0.06	2.16 ± 0.01	3.40 ± 0.01	43.25
ALMM	7.46 ± 2.12	—	—	**0.03 ± 0.01**	**0.05 ± 0.01**	30.78
MESMA	4.19 ± 0.26	3.10 ± 0.13	4.00 ± 0.22	2.32 ± 0.04	3.65 ± 0.07	36.34
NCM	5.43	2.18	3.36	2.00	3.47	40.76
GMM	3.52	**1.86**	**2.90**	2.31	3.64	65.66
DeepGUn	3.65 ± 0.07	2.72 ± 0.02	3.23 ± 0.02	2.49 ± 0.01	3.81 ± 0.01	73.33
PGMSU	**2.86 ± 0.25**	2.05 ± 0.12	3.14 ± 0.18	2.39 ± 0.02	3.70 ± 0.02	41.93

dataset. In Fig. 6, we see that the model has the best performance when J is set to 4.

The quantitative performance comparison is shown in Table II in terms of aRMSE and aSAD for abundance maps, endmembers, and reconstruction errors of the pixels. VCA suffers from outliers, observation noise, and from the random projections during processing, resulting in extracted endmembers which can vary greatly across runs. This leads to a high variance of unmixing results when combined with FCLSU as the baseline [29], [38].

Among the non-probabilistic models, FCLSU and ALMM provide poor abundance estimates due to the spectral variability, without appropriate regularization. SCLSU and ELMM give more promising results, because they promote smooth abundance maps and scaling factors. GLMM considers scaling factors for each endmember and at each wavelength and ALMM takes into account an additional dictionary to solve the mismodeling problem, which allows them to reconstruct the original pixels better, but the abundance estimates are not as good as when using SCLSU and ELMM. ALMM is a hyperparameter-sensitive model [29], resulting in unstable unmixing results under various reference endmembers extracted VCA [16]. MESMA provides satisfactory abundance estimates, but it is not practical as a time-consuming method for large images. Besides, PLMM gives the most accurate endmember estimates among the non-probabilistic models. On the other hand, the classical distribution-based models, that is, NCM and GMM, tend to absorb additional mixed pixels as endmembers leading to the saturation of abundance maps [56], and similar results occur in our experiment as shown in Fig. 5. Note that the NCM and GMM are initialized by averaging the results of multiple kMeans, so the results obtained are very repeatable once the hyperparameters are selected, as shown in Table II. The proposed method provides the lowest aRMSE of abundances of all compared methods, bringing around 18.8% performance improvement in terms of aRMSE$_{\boldsymbol{A}}$ over the second best model, GMM. The abundance maps obtained with the different methods are depicted in Fig. 5.

Visual comparison of the estimated endmember for all the methods are presented via their projection onto a 2-D space

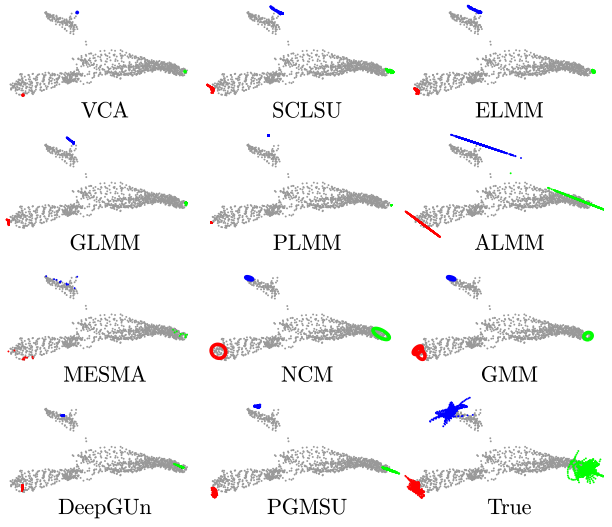


Fig. 7. Scatter plot of the synthetic dataset and estimated endmembers in 2-D space projected by PCA. Gray dots represent the pixel data and the color dots or ellipses represent the estimated endmembers on each pixel. The ellipses represent the twice standard deviation of Gaussian components for NCM and GMM.

in Fig. 7. This visualization corresponds to the geometric representation of Fig. 2. FCLSU represents the endmembers extracted by VCA. PLMM estimates a reduced diversity of endmembers because it constrains the endmembers to be close to the reference endmembers, while the estimates of SCLSU, ELMM, and GLMM are more spread. Note that there are no explicit endmembers on each pixel for ALMM, we only plot the reference endmembers weighted by scaling factors. The endmembers referred to by MESMA are extracted by bundle extraction algorithms [54], which usually choose pseudo-pure pixels that are close to the reference endmembers in SAD as endmembers.

Non-probabilistic models generally cannot capture the rich/complex variability patterns, limited by the isotropy of the Gaussian distribution and the number of components. In GMM [34], an algorithm was proposed to re-estimate

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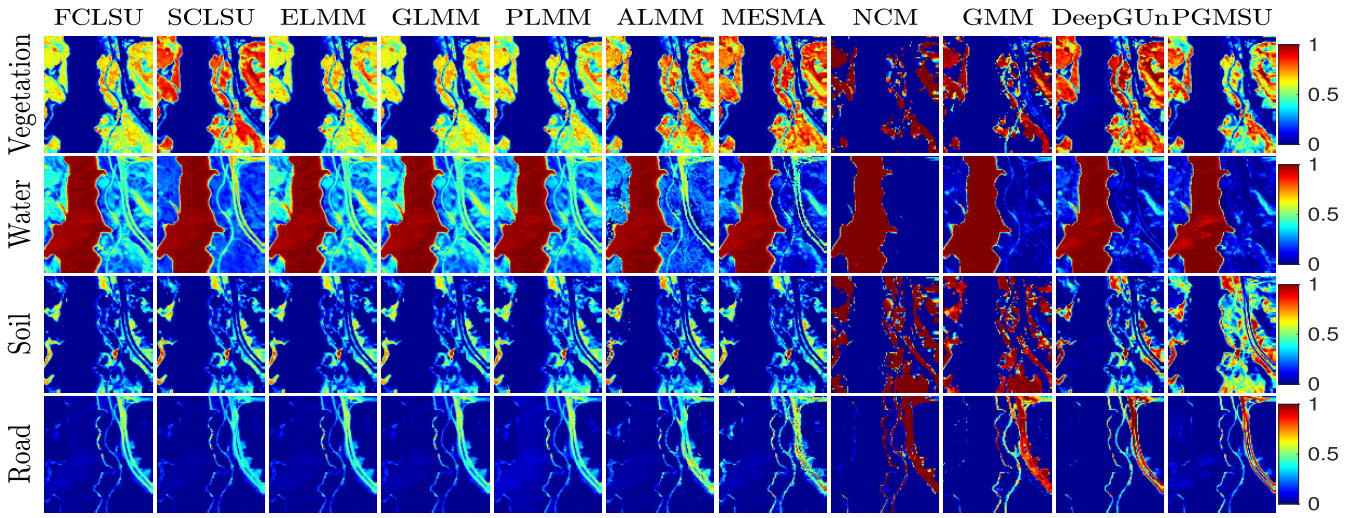


Fig. 8. Abundance maps of unmixing results for different methods on Jasper Ridge data.

TABLE III

RECONSTRUCTION ERRORS OF THE TESTED METHODS ON THE JASPER RIDGE DATASET. THE STANDARD DEVIATIONS OF ALL INDICATORS ARE THE RESULTS OF TEN INDEPENDENT RUNS. THE BEST ONE IS SHOWN IN BOLD, AND THE SECOND BEST ONE IS SHOWN IN BLUE. RUN-TIME (IN SECONDS) IS THE AVERAGE ELAPSED TIME OF EACH METHOD

Methods	aRMSE$_{\boldsymbol{Y}}$ (10^{-2})	aSAD$_{\boldsymbol{Y}}$ (10^{-2})	Run-time(s)
VCA + FCLSU	1.58 $\pm$ 0.75	10.64 $\pm$ 4.45	2.13
SCLSU	0.50 $\pm$ 0.06	3.41 $\pm$ 0.87	67.85
ELMM	0.49 $\pm$ 0.05	3.25 $\pm$ 0.60	66.85
GLMM	**0.06 $\pm$ 0.02**	**0.82 $\pm$ 0.44**	69.67
PLMM	0.65 $\pm$ 0.05	4.10 $\pm$ 0.73	407.34
ALMM	0.16 $\pm$ 0.06	1.96 $\pm$ 1.02	113.99
MESMA	1.15 $\pm$ 0.28	9.05 $\pm$ 3.74	296.43
NCM	2.39	3.09	1048.80
GMM	2.23	5.50	1786.01
DeepGUn	2.40 $\pm$ 0.33	11.07 $\pm$ 3.73	123.53
PGMSU	1.54 $\pm$ 0.03	11.11 $\pm$ 3.21	42.20

the endmembers for each pixel according to the estimated abundances and endmember distribution, so the GMM gives better endmembers estimates as shown in Table II. DeepGUn restricts the distance between the endmembers in each pixel and the reference endmembers in the latent space, which reduces the diversity of endmembers, while our proposed model is more flexible.

The elapsed times, at the rightmost column of Table II, indicate that the computational complexity of PGMSU is comparable to non-deep learning-based methods, such as GLMM, PLMM, and NCM. It is worth noting that the running time of PGMSU is shorter compared to DeepGUn as PGMSU goes further in end-to-end learning. Hence, PGMSU yielded superior unmixing performance and at a reasonable computational cost.

B. Experiments on the Jasper Ridge Dataset

To further demonstrate the benefits of our algorithm, we considered the Jasper Ridge dataset for the experiment

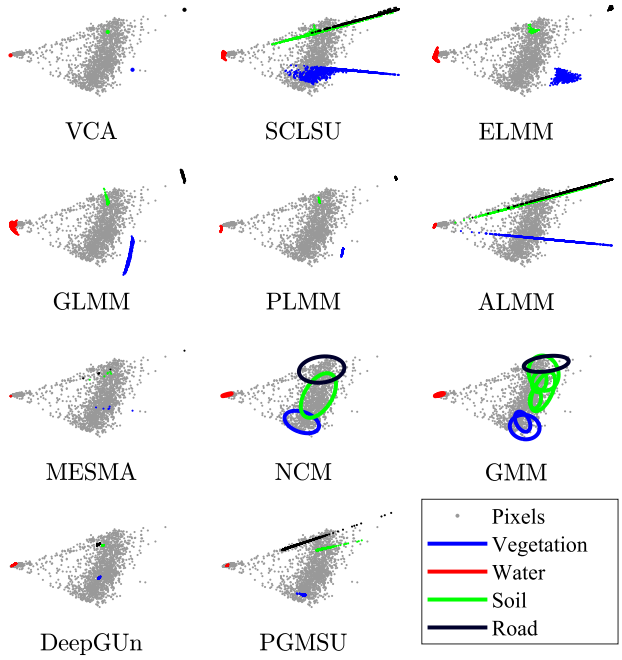


Fig. 9. Scatter plot of the Jasper Ridge dataset and estimated endmembers in 2-D space projected by PCA. The color dots and ellipses have the same meaning as in Fig. 7.

with real-world data. This dataset is popular in the unmixing task. Each pixel is recorded at 224 channels ranging from 380 nm to 2500 nm. Usually, pre-processed data are adopted, by selecting ROI and the channels 1–3, 108–112, 154–166, and 220–224 have been removed due to the water absorption and atmospheric effects. The residual bands are commonly considered to have an acceptable SNR.

The data we used contains 100 by 100 pixels with 198 bands. There are four substances in the image, including vegetation, water, soil, and road. For real datasets, hyperparameters of all algorithms are searched with the reference abundance maps[1] obtained by FCLSU with the manually

[1]https://rslab.ut.ac.ir/data

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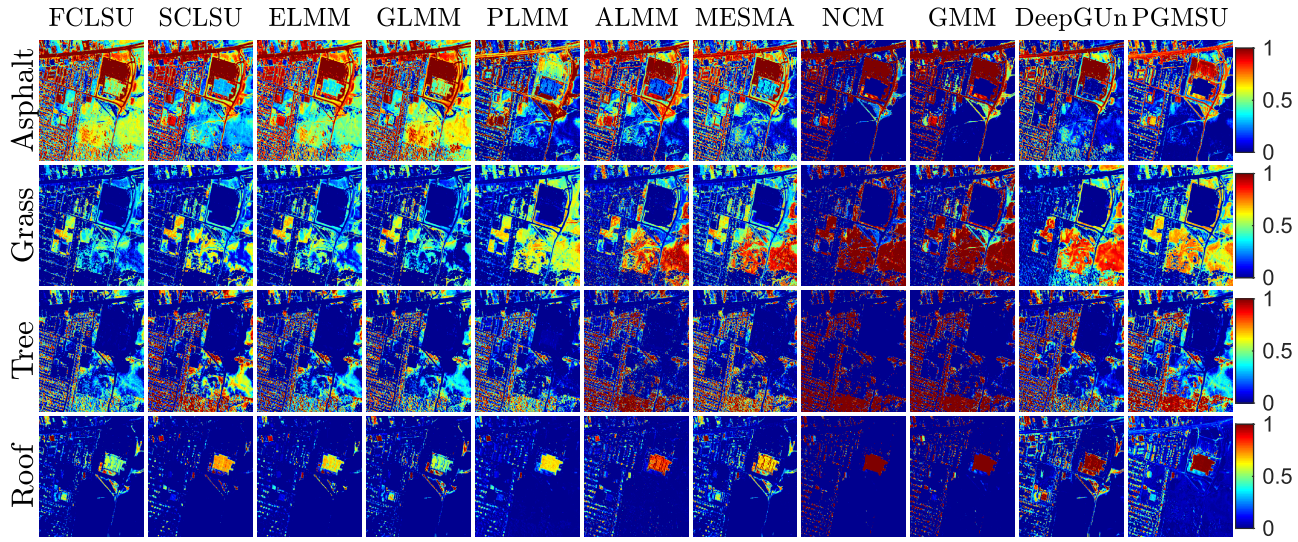


Fig. 10. Abundance maps of unmixing results for different methods on Urban dataset.

TABLE IV
RECONSTRUCTION ERRORS OF THE TESTED METHODS ON THE URBAN DATASET. THE STANDARD DEVIATIONS OF ALL INDICATORS ARE THE RESULTS OF TEN INDEPENDENT RUNS. THE BEST ONE IS SHOWN IN BOLD, AND THE SECOND BEST ONE IS SHOWN IN BLUE. RUN-TIME (IN SECONDS) IS THE AVERAGE ELAPSED TIME OF EACH METHOD

Methods	aRMSE$_{\boldsymbol{Y}}$ (10^{-2})	aSAD$_{\boldsymbol{Y}}$ (10^{-2})	Run-time(s)
VCA + FCLSU	7.36 ± 1.22	20.63 ± 7.45	21.31
SCLSU	0.50 ± 0.07	4.76 ± 1.23	234.44
ELMM	0.77 ± 0.06	4.38 ± 1.02	223.31
GLMM	**0.06 ± 0.02**	**0.32 ± 0.14**	215.78
PLMM	0.45 ± 0.05	3.10 ± 0.58	2705.63
ALMM	0.30 ± 0.11	2.16 ± 1.13	979.07
MESMA	3.28 ± 0.43	13.17 ± 4.84	350.33
NCM	7.75	16.04	2145.80
GMM	8.23	16.13	2783.11
DeepGUn	5.61 ± 0.45	17.83 ± 2.73	893.35
PGMSU	1.27 ± 0.02	6.93 ± 1.21	153.64

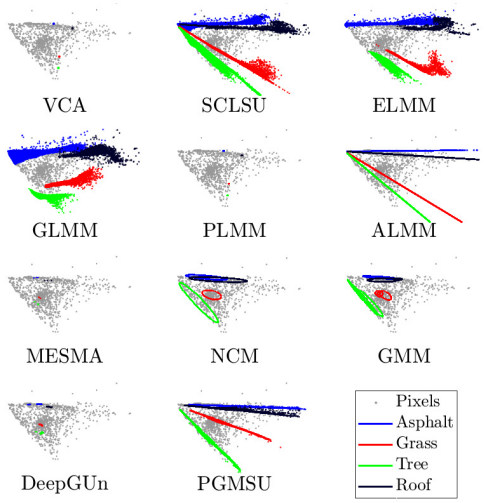


Fig. 11. Scatter plot of the Urban dataset and estimated endmembers in 2-D space projected by PCA.

marked endmembers [57]. According to the reference abundance maps, we chose hyperparameters $\lambda_1 = 0.1$, $\lambda_2 = 3$, and $\lambda_3 = 7$. Note that the so-called ground truth of these data is not actual as it was manually calculated using LMM without considering possible endmember variability and nonlinear scattering. Fine-tuning the hyperparameters on the reference abundance maps for real-world dataset will result in suboptimal hyperparameters and lead to inaccurate quantitative evaluations, for example, aRMSE$_{\boldsymbol{A}}$, aRMSE$_{\boldsymbol{M}}$, and aSAD$_{\boldsymbol{M}}$. We only evaluate the reconstruction errors as follows.

Reconstruction error as a reference is listed in Table III for evaluating the unmixing result. In a similar fashion to the results using the synthetic dataset, the ALMM and PLMM yield smaller reconstruction errors. However, there is no direct correlation between reconstruction error and abundance estimation as shown in Table II. In addition, the running time of PGMSU on the real dataset is essentially the same as on the synthetic dataset, thanks to the parallel architecture enabled by GPU implementation. Visual comparison of abundance maps with competing methods is provided in Fig. 8. Visual comparison of the estimated endmembers for all the comparison methods are shown in 2-D space in Fig. 9. Results for DeepGUn and our proposed model are more reasonable.

C. Experiments on the Urban Dataset

We further tested the performance of the proposed model on the Urban dataset. Urban is another widely used hyperspectral data for the hyperspectral unmixing issue. In this image, there are 210 wavelengths ranging from 400 nm to 2500 nm, resulting in a spectral resolution of 10 nm. To obtain data with an acceptable SNR, the channels 1–4, 76, 87, 101–111,

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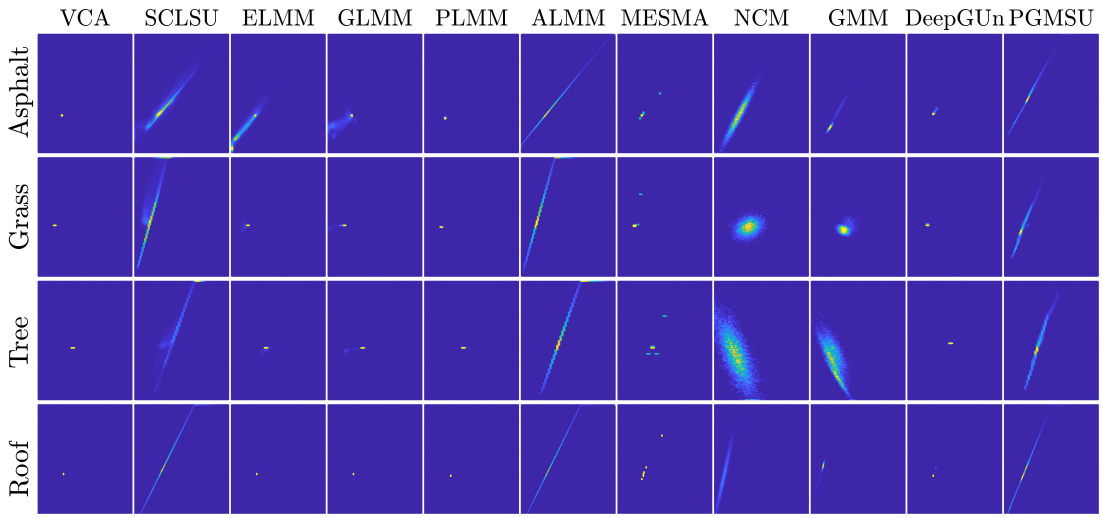


Fig. 12. 2-D histogram of estimated endmembers (band #1,#81) of the Urban dataset.

136-153, and 198–210 are removed. The data we used contains 307 by 307 pixels with 162 bands. For this dataset, we set the hyperparameters using $\lambda_1 = 0.001, \lambda_2 = 4, \lambda_3 = 6$ according to the reference abundance maps with four endmembers. Visual comparison of abundance maps with competing methods is provided in Fig. 10. The quantitative performance comparison is shown in Table IV in terms of aRMSE and aSAD for reconstruction errors of the pixels.

Note that a small value of the reconstruction error does not necessarily imply better unmixing performance. Some models may overfit to achieve a low reconstruction error, for example, the results from the synthetic data show that GLMM and ALMM resulted in high abundance errors but low reconstruction errors. Furthermore, imposing prior information on optimization variables introduces favorable properties of results meanwhile increases the reconstruction error.

Moreover, the endmember estimates of the Urban dataset obtained with the different methods are projected by PCA into 2-D space and visualized in Fig. 11. The actual endmember spectral signatures estimated by PGMSU, and we choose the band #1 and #81 of the endmember variants and compute the histogram to characterize the endmember distributions in the spectral space. Comparison results are shown in Fig. 12. Estimated endmember signatures are given in Fig. 13. One can see that there are several local maxima in the histograms, for example, GMM and PGMSU. That is to say, the endmember distributions learned by these two probabilistic-based methods are multimodal, while other methods which force the estimated endmembers to the reference ones make the results lack diversity, such as ELMM, GLMM, PLMM, and DeepGUn. Another point we want to clarify is that the PGMSU can learn not only the linear patterns of endmember distributions caused by the illumination effects, but also the nonlinear ones of spectral variants caused by intrinsic variability as shown in the last column of Fig. 12.

Finally, we discuss a possible interpretation of the estimated latent space. Four pseudo-pure tree pixels under different lighting conditions are manually extracted from the scene and

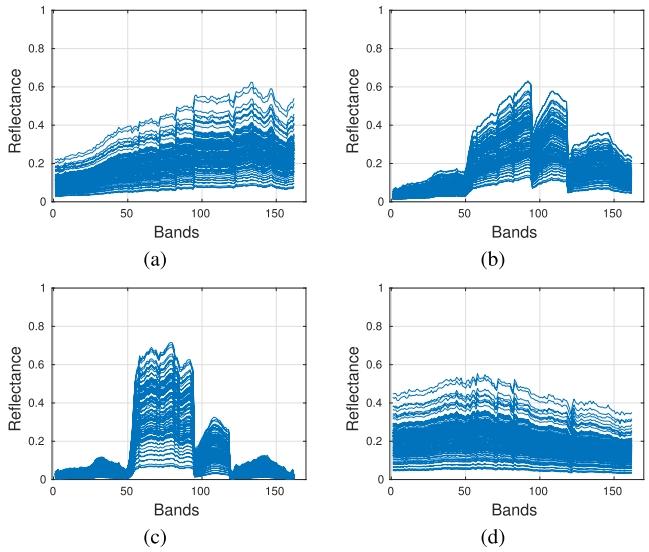


Fig. 13. Endmembers estimated by PGMSU for the Urban dataset. (a) Asphalt. (b) Grass. (c) Tree. (d) Roof.

their locations are labeled in Fig. 14(a). The corresponding spectra are shown in Fig. 14(b). Then, the inference model recognizes the distribution of latent variables corresponding to each spectrum. These distributions are projected to 2-D space by PCA for visualization, as shown in Fig. 14(c). A "lighting vector" can be found, which represents the endmember variability of the tree caused by illumination. The "lighting vector" is directed from the least illuminated pixel (purple) to the most illuminated pixel (orange) among these four pixels in the latent space. The starting and end points of the vector are the means of the two distributions, respectively, in Fig. 14(c). Furthermore, we make an interpolation on this vector and produce some new endmembers under different levels of lighting conditions via the generative model, which may not exist in the scene, as shown in Fig. 14(d). As mentioned above, there are several causes of endmember variability that are entangled in multiple dimensions of the latent variables.

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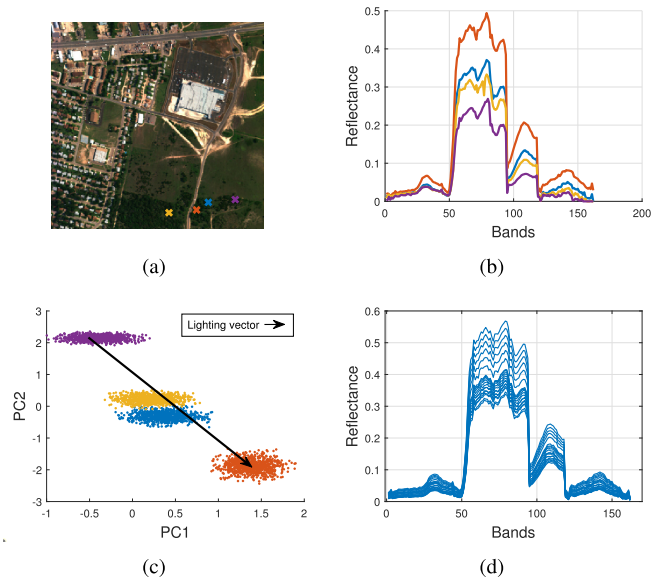


Fig. 14. Interpretation of the latent space for lighting effect for Urban dataset. (a) False color of the dataset and four pixel positions. (b) Four tree pixels. (c) Latent vectors in 2-D space projected by PCA. (d) Generated endmembers of tree.

The precise estimation and generation of endmembers in different environments require further research.

V. Conclusion

In this article, we proposed a new probabilistic generative model, referred to as PGMSU, for blind spectral unmixing. PGMSU infers latent representations of endmembers, encoding their variability, from observed pixels and it can estimate abundance maps via the inference model, simultaneously. The generative model generates endmembers for each pixel from the latent representations and then mixes them linearly using their corresponding abundances. The proposed model is able to approximate arbitrary endmember distributions, thanks to the nonlinear representation capability of the deep neural network considered. Experimental results show that our model provides competitive results and can be more effective and computationally efficient compared to other classical and state-of-the-art LMM-based models with endmember variability. Future work will focus on developing more expressive probabilistic generative models and introducing them into nonlinear unmixing to address the endmember variability issue more effectively. We also plan to disentangle the latent representation of endmembers by combining multispectral images to make the model adaptable to more complex ground environments.

Appendix A

In this section, we present detailed derivations of the loss function. For convenience, we omit the summation over pixels.

A. Derivations of the ELBO

For the ELBO, the derivations of (8) and (9) are as follows.

Using Bayes's theorem, the log-likelihood of observed pixels can be split into two terms

$$\log p(\boldsymbol{y}_n) = \log p(\boldsymbol{y}_n, \boldsymbol{z}_n) - \log p(\boldsymbol{z}_n|\boldsymbol{y}_n). \tag{27}$$

Moreover, the variational distribution, $q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)$, is introduced in order to eliminate the term related to the intractable posterior distribution $p(\boldsymbol{z}_n|\boldsymbol{y}_n)$

$$\begin{aligned}
&\text{L.H.S. of (27)}\\
&= \log \frac{p(\boldsymbol{y}_n, \boldsymbol{z}_n)}{q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)} - \log \frac{p(\boldsymbol{z}_n|\boldsymbol{y}_n)}{q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)}\\
&= \int q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n) \log \frac{p(\boldsymbol{y}_n, \boldsymbol{z}_n)}{q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)} d\boldsymbol{z}_n\\
&\quad - \int q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n) \log \frac{p(\boldsymbol{z}_n|\boldsymbol{y}_n)}{q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)} d\boldsymbol{z}_n\\
&= \mathbb{E}_{q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)}\left[\log \frac{p_{\boldsymbol{\theta}}(\boldsymbol{y}_n, \boldsymbol{z}_n)}{q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)}\right]\\
&\quad + D_{\mathrm{KL}}(q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)||p(\boldsymbol{z}_n|\boldsymbol{y}_n))
\end{aligned} \tag{28}$$

where we take expectation with respect to $\boldsymbol{z}_n$ under the distribution $q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)$ in the second equality to obtain the KL divergence. Because of the non-negativity of the KL divergence, we can drop the KL term in (28) and get a lower bound (ELBO) of the log-likelihood

$$\begin{aligned}
\log p(\boldsymbol{y}_n) &\geq \mathbb{E}_{q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)}\left[\log \frac{p_{\boldsymbol{\theta}}(\boldsymbol{y}_n, \boldsymbol{z}_n)}{q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)}\right]\\
&= \mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}_a, \boldsymbol{\phi}_z; \boldsymbol{y}_n).
\end{aligned} \tag{29}$$

Moreover, the ELBO can be expressed as

$$\begin{aligned}
&\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}_a, \boldsymbol{\phi}_z; \boldsymbol{y}_n)\\
&= \mathbb{E}_{q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)}[\log p_{\boldsymbol{\theta}}(\boldsymbol{y}_n|\boldsymbol{z}_n)]\\
&\quad + \mathbb{E}_{q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)}\left[\log \frac{p(\boldsymbol{z}_n)}{q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)}\right]\\
&= \mathbb{E}_{q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)}[\log p_{\boldsymbol{\theta}}(\boldsymbol{y}_n|\boldsymbol{z}_n)]\\
&\quad - D_{\mathrm{KL}}(q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)||p(\boldsymbol{z}_n)).
\end{aligned} \tag{30}$$

B. Derivations of the Reconstruction Error

Considering the probabilistic form of LMM (2), we have

$$\begin{aligned}
&\log p_{\boldsymbol{\theta}}(\boldsymbol{y}_n|\boldsymbol{M}_n, \boldsymbol{a}_n)\\
&\propto \frac{1}{2}\log|\boldsymbol{\Lambda}| - \frac{1}{2}(\boldsymbol{y}_n - \boldsymbol{M}_n\boldsymbol{a}_n)^{\mathrm{T}}\boldsymbol{\Lambda}^{-1}(\boldsymbol{y}_n - \boldsymbol{M}_n\boldsymbol{a}_n)\\
&= \frac{L}{2}\log\lambda_0 - \lambda_0||\boldsymbol{y}_n - \boldsymbol{M}_n\boldsymbol{a}_n||^2\\
&\propto -\lambda_0||\boldsymbol{y}_n - \boldsymbol{M}_n\boldsymbol{a}_n||^2
\end{aligned} \tag{31}$$

where λ_0 is considered a hyperparameter.

C. Derivations of the KL Divergence Error

The KL divergence in (9) can be derived as

$$\begin{aligned}
&D_{\mathrm{KL}}(q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)||p(\boldsymbol{z}_n))\\
&= \int q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)\log\frac{q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n|\boldsymbol{y}_n)}{p(\boldsymbol{z}_n)}d\boldsymbol{z}_n\\
&= \sum_{i=1}^{J}\int q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n^{(i)}|\boldsymbol{y}_n)\log\frac{q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n^{(i)}|\boldsymbol{y}_n)}{p(\boldsymbol{z}_n^{(i)})}d\boldsymbol{z}_n^{(i)}.
\end{aligned} \tag{32}$$

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In the KL divergence above, $q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n^{(i)}|\boldsymbol{y}_n)$ and $p(\boldsymbol{z}_n^{(i)})$ are both diagonal Gaussian distributions. Hence, the KL divergence can be expressed as a sum over the dimensions of the latent space. Since $p(\boldsymbol{z})$ is standard Gaussian distribution, by substituting (4) into (32), we have

$$\begin{aligned}
&\int q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n^{(i)}|\boldsymbol{y}_n)\log\frac{q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n^{(i)}|\boldsymbol{y}_n)}{p(\boldsymbol{z}^{(i)})}d\boldsymbol{z}_n^{(i)}\\
&=\int q_{\boldsymbol{\phi}_z}(\boldsymbol{z}_n^{(i)}|\boldsymbol{y}_n)\left[\log\frac{1}{\boldsymbol{\Sigma}_z^{(i,i)\frac{1}{2}}}+\frac{\boldsymbol{z}_n^{(i)2}}{2}-\frac{\left(\boldsymbol{z}_n^{(i)}-\boldsymbol{\mu}_z^{(i)}\right)^2}{2\boldsymbol{\Sigma}_z^{(i,i)}}\right]d\boldsymbol{z}_n^{(i)}\\
&=-\frac{1}{2}\left[\log\boldsymbol{\Sigma}_z^{(i,i)}-\boldsymbol{\mu}_z^{(i)2}-\boldsymbol{\Sigma}_z^{(i,i)2}+1\right]. \qquad (33)
\end{aligned}$$

By substituting (33) into (32), we obtain the expression of the KL divergence in (15).

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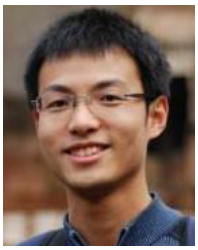
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